

Vipul Sagar

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https://vipul-sagar.vercel.app/ | Delhi, India

- **AI Tools:** ChatGPT, Claude, GitHub Copilot, OpenAI API, Groq
- **Languages:** Python, SQL, Java
- **Backend:** FastAPI
- **AI/ML:** Machine Learning, Deep Learning, NLP, LLMs, RAG, Agentic AI, Finetuning, Prompt Engineering
- **Frameworks:** LangChain, LangGraph, LangSmith, scikit-learn, TensorFlow, Keras
- **Databases:** Qdrant, FAISS, ChromaDB, Redis, PostgreSQL, MongoDB, Neo4j, MySQL
- **Analytics:** Excel, Power BI, Data Cleaning
- **Tools:** Git, Docker, Jupyter, VS Code, n8n, Supabase

PROFESSIONAL EXPERIENCE

- Web Development Intern**
Avantyx Solutions

Nov 2025 – Feb 2026
Delhi, India
- Built and deployed the official company website, optimizing UI/UX and responsive performance to drive lead generation and engagement.
 - Collaborated with the core team on GTM planning, market positioning, and digital branding during the company's foundational phase.
- Data Analytics Intern**
Indian Cybercrime Coordination Center (I4C), MHA

Jun 2025 – Aug 2025
New Delhi, India
- Designed and developed interactive dashboards for cybercrime analysis, serving 1,000+ law enforcement officials.
 - Built graph-based data models using **Neo4j** to visualize relationships between cases, suspects, and digital evidence.
 - Improved data reporting efficiency by 100%, enabling real-time insights for critical decision-making.

EDUCATION

Degree / School	Year	Score
B.Tech – AI & Data Science, Vivekananda Institute of Professional Studies (GGSIPU)	2023–2027	9 CGPA
Senior Secondary Education – CBSE, Guru Harkrishan Public School, Delhi	2022–2023	82%
Secondary Education – CBSE, Victor Public School, Delhi	2020–2021	92%

PROJECTS

- ArXiv Research Agent – Agentic RAG System** | github.com/vipulsagar25/skyclad-assignment-vipul
- Built an end-to-end agentic RAG system using LangGraph with an 8-node conditional workflow over 100+ recent cs.AI arXiv papers, enabling autonomous retrieval, clarification, web search, refusal routing, and context-aware query enrichment.
 - Improved retrieval relevance using BGE-M3 hybrid search and reranking on Qdrant, achieving a 395% retrieval improvement and increasing evaluation scores from 9/20 to 17/20 in ablation testing through optimized chunking and advanced hybrid retrieval strategies.
 - Integrated LangSmith observability for detailed per-node latency, token usage, and real-time agent workflow tracking.
- MovieGallery – Production-Grade Recommendation Engine** | github.com/vipulsagar25/MovieGallery
- Built a full-stack movie recommendation system serving 33.8M ratings across 330K users and 86K movies using ALS collaborative filtering with FAISS vector search and scalable asynchronous recommendation pipelines for personalized recommendations.
 - Architected a decoupled ML worker with 100 async coroutines, Redis-backed queues, ORJSON serialization, resilient retry handling, and background task processing, achieving sub-100ms API latency with zero failures across 134K+ HTTP requests during load testing.
 - Achieved 968 RPS with sub-4ms minimum latency using a highly optimized FAISS, Redis, and FastAPI inference pipeline.
- Clinical Co-Pilot – RAG Healthcare Assistant** | github.com/vipulsagar25/Clinical-Co-pilot
- Built end-to-end RAG pipeline using LLMs and **ChromaDB** with semantic retrieval for grounded clinical responses.
 - Implemented document chunking, embeddings, and a **FastAPI**-based backend for high-concurrency inference.
- AI Automation & Business Workflows** | github.com/vipulsagar25/N8N_workflows
- Designed conversational AI agents for Real Estate and Customer Support using **n8n** and GPT-4o with context-aware memory, automated order tracking, and dynamic lead capture workflows.
 - Built automation pipelines for cold email outreach and inventory management using Google Gemini, rule-based logic, and multi-step approval workflows.

ACHIEVEMENTS & CERTIFICATIONS

- **Smart Delhi Ideathon 2025:** 3rd Runner Up among 500+ teams, recognized by Lt. Governor of Delhi.
- **Smart India Hackathon (SIH):** 2nd Runner Up – Internal University Round.
- **HackLLM – IIIT Delhi:** Top 10 Teams among 500+ teams.
- **AI for Quality Hackathon:** Top 10 position in Rapid Fire.
- **Job Simulations:** Accenture (Data Analytics), BCG X (Gen AI) | Excel Certification (Coursera).